Cosmic Queries – Life on Other Planets with Aomawa Shields

Podcast: Startalk Podcast

AI Executive Summary

Core Thesis The episode explores the expanding frontiers of astrobiology, arguing that the search for extraterrestrial life must move beyond Earth-centric biases—specifically by investigating the habitability of M-dwarf star systems and the "terminator" zones of tidally locked planets. Through the personal memoir of Dr. Shields, the discussion posits that scientific progress is significantly enriched when interdisciplinary methods (combining astronomy with climate science and the arts) are used to foster diverse perspectives and inclusive engagement in STEM.

Key Arguments & Data * The Bias of Earth-Centric Astrobiology: Current searches for life heavily prioritize "Earth-like" planets around Sun-like (G-type) stars. However, most stars in our galaxy are much cooler, smaller, redder M-dwarfs, necessitating a broader search criteria that includes non-water or non-carbon-based life possibilities. * **Terminator Habitability:** On planets subject to tidal locking (where one side faces the star perpetually), research suggests life could potentially thrive along the "terminator"—the dividing line between the permanent day and night sides—where temperatures are more moderate. * **The Three Pillars of Life:** For a planet to be considered habitable, it generally requires three elements: a liquid solvent (typically water), an energy source (stellar radiation or chemical/thermal energy), and essential organic building blocks (such as carbon, nitrogen, phosphorus, or sulfur). * **Climate Modeling in Exoplanets:** Scientists are repurposing Earth-based climate models—originally used to predict Earth's climate change—to simulate exoplanet atmospheres by adjusting "knobs" such as stellar spectrum, orbital eccentricity, and atmospheric composition. * **Promising Solar System Candidates:** In order of probability, the most promising locations for life within our own solar system are Europa (Jupiter’s moon), Enceladus (Saturn’s moon, due to its active geysers), and potentially the subsurface of Mars. * **Diversity as a Scientific Driver:** The "Rising Star Girls" initiative argues that integrating the creative arts into science education helps girls of color develop personal connections to astronomy, which is crucial for retention in STEM fields and maximizing global intellectual capital.

Actionable Takeaways * Adopt an Interdisciplinary Mindset: To solve complex problems, look toward "cross-pollination." Applying tools from unrelated fields (e.g., using glaciology or atmospheric science in astronomy) can reveal insights that single-discipline approaches might miss. * **Challenge Foundational Biases:** When evaluating new opportunities or research areas, actively question whether your current framework is limited by "Earth-centric" or "status quo" thinking. Look for the "M-dwarf" equivalents in your own field—the overlooked variables that are actually more prevalent than the standard model. * **Integrate Diverse Identities:** For educators and professionals, do not view disparate passions (such as the arts and sciences) as conflicting. Instead, use them as complementary tools to enhance communication, creativity, and the ability to tell a compelling story.

Full Transcript

They start talking, Neil here. You're about to listen to an episode specially drawn from our archives to serve your cosmic curiosities. The archives run deep. If you enjoy this, take a peek at the full catalog on your favorite podcast platform. There's a lot there to tickle your geek underbelly. Check it out. Star Talk begins right now. This is Star Talk. Neil deGrasse Tyson here, your personal astrophysicist. I got Chuck nice with me, Chuck. Yes. Do it again. Do it again. The universe is non-stop. That's right. Is it really though, Neil? Is it really? Yes. Does it I mean seriously can we say with certainty that the universe does not stop? I mean it is anything in the universe that doesn't stop. It's the universe itself. There you go. Bam, okay I'm gonna I'm gonna see your statement and raise you entropy so Today we're talking about life on other planets. Yes life on other planets, and we've got a friend and colleague of mine So, she's an astrophysicist and more specifically an astrobiologist, which is kind of a new field. New in the last few decades, I'm old enough to call that new. Also, on the landscape of science communication, she's there and she's a force. She's an associate professor in the Department of Physics and Astronomy at UC Irvine and founder of Rising Star Girls, a program dedicated to helping girls discovered the universe. What? I know. Watch out. Also, also a trained actor. What? I know. Wait a minute. I know. I know. Wait a second. But I'm not done until I'm done. All right. Now she's got a new book out. Okay. Now you can react. Okay. What? And she's got a new book, Life on Other Planets, a memoir of finding my place in the universe.

Whoa. And that comes out in the middle of 2023. Welcome back to Star Trek. I think we've had you on before, haven't we? You have. Thank you. It's wonderful to be back. Last time we were talking about terraforming Mars, and now we get to talk about more life on other planets, places where there actually could be life, although maybe Mars. Yeah. Plus, I want to get to the bottom of what it means to have a book about life on other planets. And now you call it a memoir. Chuck, I think... She's an alien. Okay, that's what I'm saying. How was the trip? I have a lot to share. I have a lot of information now. Yes. So we'll get to your book a little later. But let me just just, can you update us on where are we in the search for life in the

universe? Because you focus on the habitability of exoplanets and in multi-planet systems.

what climate might be on those planets. So what are you doing when you go to work each day? Yes. So my team's work starts after the planets have actually been found. So it's hard enough, as you know, to find a planet around another star. And it's although we are going like gangbusters, we're approaching 6,000 planets. That's right. So, you know, and that's in the that's the tiniest fraction of our own galaxy. Never mind the fact that we have 100. billion plus galaxies in this universe, right? Yeah, we've got close to six thousand now that we've found and so Well, we don't really know anything about how habitable they might be whether they might have water on their surfaces, which is what we Define as habitable because we know all life on earth needs water everything from the tallest largest elephant to the smallest teeniest micro needs water But you just confess that you're completely biased. You're not just looking for life. You're looking for earth life. That is true. That is true. And this is why I want you to fast up just right here and now. Thank you. Show your cars. OK, go. Very true. And this is why the field of astrobiology is so important, because there are astrobiologists like me who are looking for planets that might be warm enough for liquid water on the surface, but not too warm.

And they're also astrobiologists whose job it is to ask the question, what about life as we do not know it? What about non-earth-like life? Could life use something else besides water or something else besides carbon in its backbone? So that's really important. We can't be so Earth-focused that we miss discovering life because of that. Because I've seen some missions, they're looking for Earth-like planets around sun-like stars. You can't get more sort of biased than that, it seems. True, true. And sun-like stars are not the majority of the stars in our galaxy. The majority of the stars in our galaxy are actually much cooler, smaller, and redder than the sun. So that sounds to me like we should be looking there. I have spent a lot of my career focusing on that, right? These M stars, these small, small stars. M. So we're a G star. Sun is a G star.

So with OBA of G, K, A, M. So M is like the coldest star category we have. So why would you think? Pretty cool. We know of some even cooler ones

now, like L's and T's, but then there's been some controversy. Are those really stars? Are we getting so cool that we're in the like Brown Dwarf regime? And just to be clear, when an actress is first to a cool star, we're talking a few thousand degrees. Still really hot. Yes. Not Robert Downey. You'll still be incinerated. Yeah, it's all relative. All right. So how do you modify your search for life around an M dwarf star? Yeah. If you know it's not as warm as the sun. So what happens to the habitable zone and other sort of properties that you're seeking? So if we want to look for a planet that could be what we call it, this habitable zone or the Goldilocks zone because it's not too close to its star, not too far away, so it's not too hot, not too cold. We have to look much closer to an M star than we would to around a sunlight star in the same way that if you're, you know, say you're at on the beach and you're crowding around a little campfire, you're going to have to stand much closer to that campfire to get the same amount of heat as you would if that campfire was a bonfire. The same principle exists when we talk about these cool stars versus the hotter stars. So we're really close. It just redraws your habitable zone. That's right. Yeah, so it smushes it closer up. And there's some interesting things that can happen when a planet is orbiting that close into its star. There's forces at work, there's tides, the planet pushes on the star, the star pushes on the planet, and the planet's rotation period can get slowed down and maybe even to the extreme case where there's one permanent day side and the other side it's always night and we call that situation synchronous rotation it's like this extreme case of title locking and it's literally always day on the day side always night on the night side and.

and my team. That can't be good for life. People have thought that it would be really bad. Not if you want to nap. Or maybe you just have like a work side of the planet and a party side of the planet. Oh, I like it. Just go live some in my time. You don't wait for night to come. Yeah, you go tonight. You go tonight. Oh, right. It's like, yo, what you doing later? Yo, I'm going over the night, bro. We're about, we're going to top it up crazy like. All right. It works. So that has been a big concern. However, fresh off the presses, my team actually has some new work that shows that you could have life.

along the terminator. The dividing line between the day and the night side. That's what we call the terminator. So temperature-wise, if it's always night and you're just never facing anything, that would be like a super cold part of the planet, right? Yeah. Right. And then if there was daytime, that'd be warm. That would be a super hot part, right? So there's got to be a spot. That's just right, right? Okay, and and it's funny just right spot you call the terminator Right so much more much more amenable much more neutral than our our pop fiction Schwarzenegger terminator, right? I cannot help I cannot help that we coined this term terminator habitable and that's our paper and like you can't It's like a warm bath. So before we continue, let me, if I may, I will give a brief tutorial on how stars got the letter classifications. Please. So you go back 120 years, we're getting data on stars that are not just pictures of them, but spectra. And we don't know what spectra are yet because quantum physics wasn't yet discovered.

But we knew some spectra looked like others, and you could put them on a continuum that transitioned from one kind of spectrum to another. And so we did that. And we lettered them A, B, C, D, E, F, G, H, I, G, the elemental B. Then quantum physics came around, and we said, yo, there are reasons why the spectra looked this way, and it's not what you're thinking. And it had to do with the temperature of the star. And it turns out these features can duplicate between being too cool to have a feature and too warm to have that same feature. By the time we shuffled these letters into a temperature sequence, A, B, C, D, E, F, G became O, B, A, F, G, K, M. And why didn't we just re-letter them, which I think we should have at the time? But we didn't. So we have this artifact. Could have gone with numbers. Pretty simple. Yeah. Because now we get to confuse astronomy students the world over year after year. Forever and ever and ever. Every year I get that question. So both stars are hot. B stars are next hoted. O, B, A, G, we're G star. And then each category split into 10 other subcategories, okay, zero through nine. So we are the G2. Oh. Okay, Arabic numeral two, okay. And we divided them by what's called luminosity class. Like how big are you?

And so we have classes one through five or four or three. Yeah, five. And so we are G Roman numeral two. I've got G Arabic numeral two, Roman numeral five. Wow. You tell that to an astrophysicist, they know exactly what kind of star you're talking about. G to five. That's what it is. Sounds

like something that rappers fly to their next concert in. G to five, baby. So I don't know why. Tell me, what did you need to do to model a terminator habitability? What kind of calculations were necessary there? Yes, and I have to give a shout out to my post doc, Anna Lobo, who's spearheaded this work. And we had to ask that question, as I write in the book, that having this question, could these planets even exist? It's the first place to start for any...

scientists or any curious person. Before you invent life on a planet, make sure that kind of planet could actually exist. And the first thing we had to do, we're using models, climate models that are normally used to predict the climate on Earth. So they're the same models that have been used to predict the effects of climate change into the 2100s. The reason why we know that climate change is real and that it's going to continue unless we do something is in part because we're using these models to forecast those effects. But what we're doing... And these models, so all you have to do is change some of the... They probably have knobs you can turn. That's right. Everything else, because the physics would just be the physics, but the rotation ray and the sun's energy entering the system.

These are knobs, right? These are knobs that we can turn. I didn't know these are general climate models. Fascinating. That's right. So we are changing the spectrum of the host star, how its energy, its light is distributed across the whole spectrum is going to be different depending on the temperature of the star, the type of star. We're changing the shape of its orbit, how much is in it, how much atmosphere, what its atmospheric composition is. because we don't know those things for Earth-sized planets. We have no information about the type of atmospheres they might have or their surfaces. And understanding the effects on climate of different types of atmospheres is going to help us understand which of these nearly 6,000 planets we want to prioritize and look at with next large generation telescopes to find out if there really is life there.

So this is a beautiful example of the cross-pollination between two otherwise very different disciplines. Yes. You're going to climate science folks. They don't care about exoplanets. They just care about Earth. I know and I'm on a personal crusade to change that because it's true. It's so

interdisciplinary. We're using atmospheric science, climate science, astronomy, geology in some cases, glaciology because I'm all about ice and different types of ices. These different types of ices interact with different types of light depending on the type of ice. There's more than just water ice. We're looking at CO2 ice and other ice. So different ice would reflect their sunlight differently from other kinds of surfaces and would change how much energy enters the system. That's right. That's exactly right. I'm Nicholas Costella and I'm a proud supporter of Star Talk on Patreon. This is Star Talk with Neil deGrasse Tyson.

So Alamoa tell me what how does all this fold into this this book that you just wrote? I'm so excited for this book. This has been a dream of mine for a long time and as you know I have I was an astronomer then I trained I changed and studied acting and then I came back and finished my PhD and became a professional astronomer and a professor and For the longest time I was trying to reconcile these two things that I loved acting and astronomy and I thought

that they couldn't coexist. I thought I had to choose. And when I would share with people those two loves, I got a lot of like, what? This is so strange. Why? How? And I, you know, I think I internalized that and tried to figure it out. And it was funny because when I stopped figuring it out, these incredible miracles got to happen where I saw how they can really coexist, like hosting a science TV show, which I could never have planned on my own. Eventually going back to grad school in part because of your suggestion and being put in touch with you and your mentorship. And also I had applied to the Astronaut Canada program and had not reached the next level. It was like everything was saying, go back, get your PhD. And I finally became willing to listen. And so the book really is about that whole journey. It's about why I fell in love with the stars and the night sky.

and why I fell in love with acting and this whole edge that I carried for a long time of like feeling like I was an astronomer or a scientist around actors, feeling like I was an actor around scientists and eventually coming into full acceptance of this non-traditional background that I had and how it could really make me a better scientist and a better communicator because of it. It's about the whole journey. First of all, that's a beautiful what you

just said because it's a reconciliation of self, which I think is always a wonderful thing that anyone has ever... You could be a counselor. You could be like a therapist. Yeah, if you want to mess up your life. No, I think if you have these loves...

I had two heads of secondary pathways, but none as significant in my life as your acting was. But I just did little things. I doodled, and I liked calligraphy when I should have been doing other things. But now when I sign my books, I use a calligraphic pen. And the people who received the book, they appreciated that much. It was little things like that. You find a way to fold it in. Your life and the lives of others are much more enriched. So I'm so glad that you pursued what you love rather than what people thought you should be doing. Thank you. I I have to say I still remember and you're gonna find out you you probably read parts of it already But if not, you'll see later like you're in the book Wait a minute. Am I in the book? You're gonna have to read it to find out And you're in the book There are a lot of people are in the book But like i write about that moment when you know i first met you in person at the double as the american astronomical society that's our professional conference for for those who don't know what double as means and. We sat you know on some umbrella chairs or whatever and people were walking by like is that. How do we think it is is that neil degrasse ties and you know and and you are just there and i think i like.

Started to cry at one point because I had you know, we've been talking about my background and how I had no I had there were ups and downs I had struggled academically because my head was not in the place like my my heart and my head were not in the same place You know, you know, it's funny because I cried when I met Neil too But but that's because he was standing on my foot the whole time And he's a big guy He's a large man. It was not pleasant. Wait, Omoa, plus it didn't help matters that you also married an actor. Isn't that right? If I remember correctly, is that correct? I did, yes. He is 100% actor. He doesn't have the, like, actor, scientist, engineer, architect. Distraction, distraction. He's 100% actor, although he does enjoy those, like, doomsday.

Nat Geo shows or like Super Volcano or, you know. Okay, yeah. And he's a lot. But what I'm saying is that that would have been a force that would have been supportive of your acting, correct? Am I presuming? Absolutely. We met in acting grad school. He was my classmate. Oh, God, you are becoming a real actor. She said acting grad school. Yes. Okay. Listen. I, yeah, I. That's the real stuff. That's when you're just like, you know how people say, well, I'm going to be an actor. And they're like, well, you're never going to get a job. When you say, I'm going to acting grad school, they'll be like, oh, you're getting a job. That's what we hoped and expected. That wasn't always the case. But yes, I wanted to study it in the same way that I studied astronomy. And that was so much fun.

So I had left the PhD program that I was struggling academically in and I had an old white male professor tell me to consider other career options and, you know, and I internalized that and left. And, you know, when I got to acting grad school, it was like I was free. I felt so light. But it wasn't like it was easier compared to astronomy grad school. It was harder in a different way. And I write about this. It wasn't necessarily about the brain cells working over time. It was about dredging up my emotions and feelings and childhood experiences and bringing it all up. That I needed to use it to embody these different characters. Summoning it on command. Summoning it on command, yeah. It's gotta be real. It's gotta be real. So you stitched this. You made a tapestry of your life with your professional ambitions.

in this book. So it's part memoir, part advice column. Is that fair? There's certainly, it's all spoken for the most part in the first person. Like this is what I did. This is what helped me make my life better and reconcile these different parts of myself. And I hope that I'm writing it for the people that have that question of like, is it too late? There's that part of me that I never could bring up into my life. but it's always been there like I say in the book it's when you leave a dream behind it never dies it's like sitting on the side of the road and eventually it'll catch up and that's what happened for me you know and and the whole message of the book is that it's not too late and that if there's no role model that's doing what you want to do you can be your own role model yeah yeah that's right and in fact if you always needed a role model you would only ever do things that other people did before you. But sometimes you have to failblaze. And like you just said, if you become

your own role model, that's harder, of course. And you got to figure things out on the fly. But if and when you succeed, then people say, of course, that's what I always wanted to do. Those who come behind you. So just congratulations. Yeah, thank you. It was fantastic. Well, Chuck, let's get to some queries. All right, why not? Especially tune for this visit. Absolutely.

These are questions from our Patreon members? Patreon patrons who just for a haul tree almost insignificant five dollars a month. All right, this is Mike Parker and Mike Parker says, hello Dr. Shields. Mike Parker here from Richmond, Virginia. If life does exist in our solar system, which planet or moon do you think offers the most promising location. I love it. In fact, give me the top three cases in order. Oh my gosh. That's just one. Okay. This is a fantastic question. Thank you for asking it. Europa, which is Jupiter, a moon of Jupiter, would be my top. Even though there's a lot of buzz right now around Saturn's moon Enceladus, which would be my number two, the next one. Saturn's moon Enceladus has geysers.

Geysers of liquid water that are shooting out from its south pole and there was a reason and can't argue with geysers And apparently all of the basic elements of life are there Geysers lead to geysers Geysers lead to geysers I don't think that's how language works Okay And third one give me a third one I'm thinking I mean I I'm skeptical about Mars. Mars. Under-surface Mars. Under-surface. You know there might be some liquid water in aquifers. Yeah. I'll be your third... I was about to say that, but I'm skeptical. Or it could be Mars might have fossilized life. Okay. So the life was there, it's no longer there, but that still counts. Because it ain't here if we can prove that it was there, it still counts as life because it was there. True.

I'll give you that. And there's so much evidence of liquid water on the surface at some point. At one point, yeah. All right. Excellent. Excellent question there. Nice question. Keep it going, Chuck. Mike Parker, way to go. All right, here we go. This is Sherry Carisou. And Sherry Carisou says, hello, Dr. Shields. Do the ideal conditions for life on extrasolar? planets depend on the star itself, or are the requirements the same no matter what? Also, thank you, thank you for Rising Star Girls. I am registering my niece. You are an inspiration, Sherri, from San Diego. Thank you so much, Sherri,

for asking that question and can't wait to see your niece in Rising Star Girls workshops. Rising Star Girls posts summer.

We host annual virtual workshops during the summer for middle school girls of color. And that means ages 10 to 14 or 15, roughly. No one is turned away. It's virtual. And they're virtual. So we have girls that participate from all over the country. And we even have girls participate from other countries, too. And we're ramping up to do even more. We also host educator webinars where we We show educators how to use the activities in our handbook with girls in their own communities all over the world. So and and back to your question. The ingredient week the primary ingredients for life are. Liquid water or a liquid which on our planet is water and as we know we can't just look for water but that's like that's our first order thing we're looking for some liquid that that.

Life can use to make it's the stuff it needs to carry out its chemical processes and reactions some kind of energy source whether it's the Sun the star or Chemical energy like life that lives in the deep ocean around these hydrothermal vents has no access to sunlight and yet there's life there So it's using the chemistry of rocks and heat from from the core of the earth and I add there that when I was taught biology It was only assumed that life could thrive. It was assumed that life could only use sunlight because we hadn't discovered the life at the bottom of the ocean yet. So they had to broaden the definition. Not that life depends on the sun, but life depends on an energy source. So I like this, that that's now a folded into that definition. Okay, so keep going. And then the last one is the basic building blocks of life. Like some kind of environment for life to make its...

know to carry out its metabolic processes and to use so we think of this like it needs sulfur or phosphorus or oxygen or nitrogen or carbon. These are like the basic elements of life. So. The organic chemistry. Yeah, the organics, organics, energy source and liquid. So all of those things are what we look for when we're looking for a planet that could be habitable. And for the part that I am really focused on is where could the climate be suitable for water to not freeze, not evaporate away, but to stay in liquid form somewhere on that surface. And so this can be affected by the type of star as you bring up in your question. Because the starlight can be different

depending on the temperature of the star, how that starlight interacts with the surface of the planet.

the various atmospheric molecules on that planet, that can influence the planet's climate. So it does depend on the star. But those three ingredients are the main ingredients that we think any kind of planet would need to host life, to keep life going for a long time. Now you just made me think of a question that if it's a dumb question, just let me know. There are no dumb questions. That is not true. You have not heard me ask questions. When people say there's no dumb questions, I'm like, just wait. No, but so are there different stars that give off different colors and can look like our stars white? Are there different stars that give off different colors? And can those different colors actually affect the development of life? That is the complete opposite of a dumb question.

That is a fantastic question. So yeah, I'm not going to believe anything you say now. OK. It's not a tough question. It just popped in my head. So when it just pops in my head, I'm like, maybe that's a dumbass question. Never, never. So yes, different stars have different colors. And if you if you are fortunate enough to live someplace where you don't have a lot of light pollution, you go out on your porch or your stoop and look up, you'll see this. You'll see this in action that. All of the stars don't all look the same. Some of them look white, some of them look yellow, some look a little orange or a little red. And if they're twinkling, that's a pretty good bet that that's a star. If it's not twinkling, then you can be more convinced that it's a planet. There's a lot more stuff between us and a star.

than there is between us and a planet in our own solar system. And so there's a lot more atmospheric distortion and turbulence, and that's what we see as twinkling. But anyway, yes, the colors are real, and that has to do with the temperature of the star and how much light it's emitting. And that light, again, that's gonna, if there's a planet around that star, and just about every star in our galaxy has a planet around it. may not be in the habitable zone, but if just about every star has a planet, then that planet, if it's got an atmosphere, if it's got a surface, that light is going to be shining down on that planet. And how much light is in what region of the spectrum, the color of that light is definitely going to influence how it interacts with what's on

that planet and the weather, the temperature, the climate of that planet. So you have one of the most complicated problems to solve out there, it seems to me.

Because it's not just light shining the planet. It's like, how does the chemistry of the atmosphere interact with the light that's shining down on the planet? Does it get absorbed? Does it get readmitted? Is it reflected? And then if it reaches the surface, then you got to worry about the surface. And in some way in there, maybe there's life. It sounds like a really hard problem. It is a really hard problem. Because those same cool stars that comprise 70% of all stars in the galaxy, those M stars, they're really awesome because they're so numerous. And there's a lot of advantages to looking for life around those stars, but there's some disadvantages, too. They have really long phases where they send out a lot of X-ray and UV radiation shooting towards the surface. And we know the reason why we wear sunscreen is that UV radiation is not good for biology. And so it could be that life could only exist at the bottom of the ocean on those planets.

where you're protected from the UV. Water's a good UV absorber. Super cool. All right, keep it coming. All right, here we go. This is Omar Marchelino and Omar says... Hello, Dr. Tyson. Hello, Chuck and Dr. Shields. Omar here from Dallas, Texas. Dr. Shields, giving your unique journey through the cosmos of astronomy and theater, two fields that are often mentioned never in the same sentence, let alone in the same career. You have boldly gone where few have gone before and leaving the way and ensuring others from diverse backgrounds can do the same through your work with your rising star girls. In your perspective, how could this universe of diverse talent and experiences propel us further into the final frontier. How do you think incorporating a spectrum of perspectives in astronomy could accelerate our interstellar mission to understand and reach other worlds?

Excellently written question and and recited so yeah, I'm gonna say Omar. You want to read this book? This has got the whole journey in there pretty much everything what you did there and go to risingstargirls.org because you'll you'll see We're not just Teaching these girls astronomy and astrobiology. We're not just saying hey, there's stars out there. There's

galaxies. These are the names of the they get to write a poem about what they're learning, they get to design their own exoplanet and make choices about how many stars it orbits, and if there's life there, and if there's not, why not? They get to write their own constellation myths. They get to calculate the distance from the Earth to different solar system bodies and units of themselves. So we're doing this all to develop a personal connection between these girls.

and the universe of which they're an integral part. So we're processing what they learn through a creative arts-based lens. And I wanted to bring it back to Rising Star Girls because that's really, my journey has been epitomized by this program of like, they're not these, the arts and the sciences are not as disparate as many people might think. In both cases, it's about the story, right? Everything has a story. Planets, stars, they all live, right? And they die. And there's the story of their becoming, and there's a story of their evolution, and then there's a story of their deaths. Planets don't have mother issues, right? Later in life. That's right. There's certain stories they're not going to tell. I'm pretty sure it's the stories. The stories are different, but they, but there's stories nonetheless, or the stories of humans on the stage or on the screen, right? Looking to live as, as one of my method acting teachers used to say, right? Trying to find happiness and fulfillment. So there's that whole like science in the arts. And in fact, not only are they not as disparate as we might think, but they actually interweaving them is the key in my mind to answering your question is to creating imaginative thinkers and enlightened scientists and enlightened artists for that matter, realizing that these disciplines can fuel each other, can be used to elevate our core understanding of the human experience and the universal experience. So that would be my best way to to answer that really exciting question. Can I answer that as a minimum in a day when only men did science? It means half the population of the world, half the intellectual capital of our species, was not at the table at the time anyone is thinking about how the universe works. So the extent to which you can extend that to have 100% of our species have access to whether they choose it, that's another question.

access to it, then you can ensure that we're getting the best minds who want to actually think about our past, present, and future in the universe. And

right now, that is not the case. That is absolutely not the case. Physically, it was half. Percentage-wise, it was way more than half. Okay, it was half of the mass, but in less than half the actual brain power. There you go. I get it. Yeah, and we want these girls to claim ownership of what they're learning, you know, to know that they matter, that they're an integral part of this universe, that, you know, just in the same way that if we were to look at the Milky Way galaxy through one type of light, just visible light, you'd see like this thin This thin line with a bunch of dark clouds in it, and you'd think there was nothing there. If all you looked at, if you looked towards the center of the Milky Way galaxy, that swath of space that some of us can see if we go out in the desert, if you just thought that that was the only... All of us could see. Yeah. Remaining 3% of the population of the world. Right, where there's no extra lights pollution.

If we thought there was nothing there, we'd miss out on so much information about this galaxy. But if we looked at the infrared, we see so much going on. Never mind UV and gamma, there's so much more to the galaxy than meets the eye. So too is there's so much more to these girls than meets the eye. We want them to have that ownership of that star, that planet, that galaxy that they're studying so that when they continue on and the heavy math comes in, they'd be less likely to leave the field because they think, oh, I didn't get that question right on that test. I didn't know if someone told me that I shouldn't be here. Just knowing that there's a connection that they have between what they're learning and who they are that we hope will really help them stay. Cool. Let's see if we can slip in two more questions, Jack.

All right. Well, here's one from Carrie Manneberg. And Carrie says, Hi, Dr. Shields. Is there any scientific evidence that life on Earth came here from an interaction with a comet or asteroid that had biological foundation chemicals? Oh, this is a this is a popular question within the astrobiology community. I believe the term for this is panspermia. Yes. Right. That something comes in from other from elsewhere and like seeds. Seeds us. Yeah. Yeah. I mean, of course it's possible. I don't think it's being given very much funding, to be honest, and in terms of exploration of this topic, but it certainly has been discussed within astrobiology communities. Okay.

Okay. All right. Okay. This is Jason. He says, hello, Dr. Shields. Do you have any personal moral quandaries about science moving forward?

or is pretty much just forget morals and let's just see how far we can go. Now, I don't know if Jason is making an indictment of any sort or not. That's a very specific question. It's starting out with the premise that all scientists are immoral. Yeah, basically. But I think I know what he means. It's like scientists want to discover. What if you and you two, Neil, had the opportunity to make a discovery that would enlightened, but there was a larger chance that it would destroy. Would you still expose that discovery? That's a moral quandary. Yeah, this is an important question. This harkens back to the last episode that I was fortunate enough to be on with you, you know, where we talked about terraforming and we talked about some of the moral and ethical implications of that.

that question of, is it okay to terraform another planet or terraform Mars? I do think that as scientists, ethics must be key. So I don't subscribe to the view that science and discovery at all costs. There's a piece that I've written for an upcoming periodical that's like... Why why why do this? Why look for life elsewhere? Why spend billions of dollars doing all of this and and is that object? Umu Amua that first interstellar object, you know, is that you know, is it really an alien's? Craft or is it you know, is it just can it be explained by regular? Regular scientific scientific method. I think what it comes down to is if we go And I'm talking about this in the context of the search for life elsewhere. There's all sorts of other genetics and questions about cloning and all of that that have their own ethical implications. But from my part of the sphere on that question, if we were to discover that we are not alone, what would we do with that information? Would we have to go and visit? Or could we stay on our...

planet and know that we're not alone and let that information inform how we think and feel and move throughout the world. Or would we need to go? And if we would need to go and visit, could we just go and visit and then come back home? Or would we need to stay? And if we needed to stay, could we just stay and not try to conquer? Or would we have to go the way that and emulate what was done on our planet with? Columbus and Magellan and all the rest and the implications of that, right, that led to

genocide. I would hope that we would be able to learn from the lessons of the past and not repeat them. And that really is what I think it all comes back to, is hope. I have no clear answer about what other scientists might do and maybe even what I might do if given the opportunity to go to another world and communicate with other species.

But I carry with me the hope that I would want to do whatever we do from a place of love, kindness, inclusivity, and welcome and humility, to be honest. So that, I think, is the key, that to be a scientist, I must concede that there is more that out there that can be explained. And I can hold on to that humility and help and and allow that to guide to serve as a North Star for me morally So well that is beautiful as a comedian. Let me just say this that ain't happening Not what is going to happen. Well, but I got two reactions one I how charming it was that you thought we would go find aliens on another planet and that we might exploit them It's just. Oh, rather than like Independence Day and they're going to come and kill us. Oh, yeah, yeah, yeah. And believe me, if we go see them, they're not answering the door. They're going to play the role just like we Jehovah's Witnesses or something. They'll be like, they're here. Right. Don't let them. We're not. Get away from the window. Get away from the window. I've never had heard anyone talk about an encounter with aliens and talk about we exploiting them. That is like never.

I don't know any storyline. If you're thinking of that storyline, that is completely beautiful. But of course it mirrors what happened on our planet. I mean, how could I not think of it that way? You know what I think about it? I think if they come here and they do what we do to each other, as we portray in the films, then we're creating them not based on a supposition of how they would behave, but on actual knowledge. of how we did behave. So, in fact, all of the alien movies are mirrors to our culture as you... Treating it almost like a guidebook or a recipe. Yes, yes. And I couldn't help... You have to publish a paper on the Interstellar visitor Amur Amur. So, we have Aumawa... I know, it's so close to me.

We need that research paper. I write it. I write in the book. I say umuamua. It sounds like my name. Yeah, that's great. That's right. Umuamua is Hawaiian. And does that have some origins? My parents made it up. They

made up, yeah, Val sounds, they put together their musicians, they made a chant up, but yeah, that sounds, I went to Hawaii once. They sounded like hippies. They sounded like some leftover hippies right there. All the way. Very nice, very nice. Well, at least you weren't named Moon Unit, okay? Right. Or pilot inspector, I mean, come on. That's unfortunate. But yeah, umuamua means messenger from afar arriving first. Yes, first messenger, yes, yeah. Yeah, so guys, we got to end it there. That's sad. Oh, my God. This has been fantastic. Thanks for being on it. Congratulations again. Thank you so much, Neil. And thank you, Chuck. It's great to be here. All right. Chuck, good to have you, man. Always a pleasure. All right. I'm Neil deGrasse Tyson. You're a personal astrophysicist for StarTalk. As always, I bid you to keep looking up.